6-th

5 number Summary And Boxplot

*Outliers

(i) Minimum

ii) First Quartile (25 percentile) Q1

iii) Median

iv) Third Quartile

v) Maximum

Removing the outliers

$x = \{1, 2, 2, 2, \boxed{3}, 3, 4, 5, 5, 5,$
$6, 6, 6, 7, \boxed{8}, 8, 9, 29\}$

*outlier

[lower fence $\longleftrightarrow$ Higher Fence]

Lower Fence = $Q_1 - 1.5(IQR)$

$\downarrow$

Inter quartile range

Higher Fence = $Q_3 + 1.5(IQR)$

$Q_1 = 25$ per $= \frac{25}{100} \times (19+1)$

$= \frac{25}{100} \times 20$

$Q_1 = 3$

$= 5^{th}$ value

$Q_3 = 75$ percentile $= \frac{\cancel{75}^{15}}{\cancel{100}} \times \cancel{(20)}$

$= 15^{th}$ value

$Q = 8$

$$\boxed{\therefore IQR = 8 - 3 = 5}$$

lower Fence $= Q_1 - 1.5(IQR)$

$= 3 - 1.5(5)$

$= 3 - 7.5$

$= -4.5$

High Fence $= Q_3 + 1.5(IQR)$

$= 8 + 1.5(5)$

$= 8 + 7.5$

$= 15.5$

$[-4.5, 15.5]$

5 number Summary

Min $= 1$ $\quad$ $Q_1 = 3$

Median $= 5$

$Q_3 = 8$

Box Plot (outliers)